

Total Variation of Function

and

Functions of Bounded Variations

Seminar Presentation

The question

How much does a function move?

Consider a function,

$$f : [a, b] \longrightarrow \mathbb{R}.$$

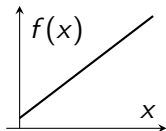
As we move from a to b , the function may,

go up go down oscillate.

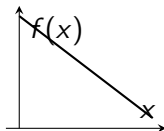
Can we quantify this movement?

Different kinds of movement

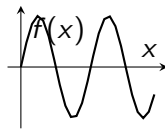
Increasing



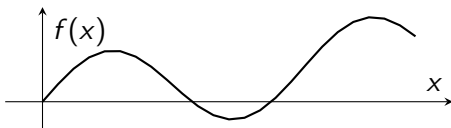
Decreasing



Oscillatory



Up and down



Net change is not the whole story

Suppose the function moves as

$$0 \longrightarrow 5 \longrightarrow 2.$$

Its net change is only

$$2 - 0 = 2.$$

But it actually moved

$$5 + 3 = 8.$$

We want total movement, not just net movement.

Looking at the function through a partition

Choose points

$$a = x_0 < x_1 < \cdots < x_k = b.$$

We only look at the function at these points:

$$f(x_0), f(x_1), \dots, f(x_k).$$

Between two consecutive points, the amount of movement is

$$\boxed{|f(x_i) - f(x_{i-1})|}.$$

Adding all these movements gives an estimate of the total movement.

Variation with respect to a partition

Let

$$P = \{x_0, x_1, \dots, x_k\}$$

be a partition of $[a, b]$.

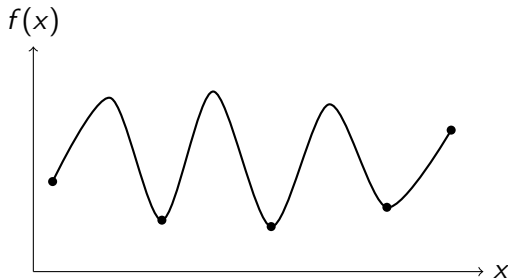
Define

$$V(f, P) = \sum_{i=1}^k |f(x_i) - f(x_{i-1})|$$

$V(f, P)$ measures the movement detected by the partition P .

Why does the partition matter?

A coarse partition may miss some of the behaviour of the function.



More points $\Rightarrow$ more movement can be detected.

From one partition to all partitions

The value $V(f, P)$ depends on the choice of P .

So we ask:

What is the largest variation obtainable from all possible partitions?

This leads naturally to

$$TV(f) = \sup_P V(f, P)$$

where P ranges over all partitions of $[a, b]$.

Total variation

$$TV(f) = \sup \{ V(f, P) \mid P \text{ is a partition of } [a, b] \}$$

Conceptually:

one partition $\longrightarrow$ all partitions $\longrightarrow$ total variation

Bounded variation

We now have a natural question:

Can the total movement be infinite?

A function f is said to be of **bounded variation** on $[a, b]$ if

$$TV(f) < \infty.$$

In other words,

the function cannot accumulate arbitrarily large total movement on the interval.

Example: an increasing function

Suppose f is increasing on $[a, b]$.

For any partition

$$a = x_0 < x_1 < \cdots < x_k = b,$$

we have

$$f(x_i) - f(x_{i-1}) \geq 0.$$

Therefore

$$\begin{aligned} V(f, P) &= \sum_{i=1}^k |f(x_i) - f(x_{i-1})| \\ &= \sum_{i=1}^k (f(x_i) - f(x_{i-1})). \end{aligned}$$

The sum telescopes

$$\begin{aligned} V(f, P) &= [f(x_1) - f(x_0)] + [f(x_2) - f(x_1)] + \cdots \\ &\quad + [f(x_k) - f(x_{k-1})] \\ &= f(x_k) - f(x_0) \\ &= f(b) - f(a). \end{aligned}$$

Remarkably, this is true for **every partition** P .

Therefore...

Since

$$V(f, P) = f(b) - f(a)$$

for every partition P ,

$$\boxed{TV(f) = f(b) - f(a)}$$

for an increasing function.

For a monotone function, there is no wasted movement.

The picture behind the definition

Change $\longrightarrow$ Movement $\longrightarrow$ Variation

Variation over one partition $\longrightarrow$ Total variation

$TV(f) < \infty \implies f$ has bounded variation